



Society of Petroleum Engineers

SPE-229612-MS

Innovative Execution of a Complex Rigless Well Intervention in an ERD Well Using Slim Tractor BHA and High Mast in Cambay Offshore Field

Jeevan Jyothi Chakka, Anmol Chhoker, and Vishal Ranjan, Cairn Oil & Gas, Vedanta Ltd, Surat, Gujarat, India

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229612-MS](https://doi.org/10.2118/229612-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper details the successful planning and execution of a complex rigless well intervention in an Extended Reach Drilling (ERD) well located in the Cambay offshore field, India. The objective was to diagnose the root cause of high water cut and restore oil production by isolating water-producing zones using advanced production logging and mechanical intervention tools, despite significant operational and infrastructure challenges. The subject well is a high-angle, multi-zone horizontal ERD well with a measured depth exceeding 3,600m and deviation approaching 82°. Following a sharp increase in water cut (>95%), initial diagnostic and conveyance attempts using conventional tools were unsuccessful due to internal completion restrictions and the well's challenging trajectory. The intervention team adopted a custom-engineered 2-1/2-in. slim-profile electric-line tractor designed to navigate through the well specific restrictions and tractored successfully in a 5-1/2-in. lower completion. The Multi Array production logging tool, featuring array of tools, was specifically selected due to the high deviation of the well, where conventional single sensor tools typically fail to provide reliable results. This configuration allowed for precise fluid profiling and accurate detection of segregated multiphase flow in the horizontal section. Due to rig up height constraints on the offshore platform, a 90-ft portable mast—the first of its kind in the Cambay Basin—was deployed and validated by third party structural studies & engineering evaluations. Detailed risk assessments and weather-specific contingency plans ensured safe operations.

Based on production logging interpretation, two sleeves contributing high water influx were selectively closed using a 2-1/8-in. high- expansion stroker. Post-intervention, the well achieved a stabilized oil production of ~590 BOPD with a reduced water cut of 75%, with zero HSE incidents or Non-Productive Time (NPT).

Novel/Additive Information: This case marks the first successful integration of slim e-line tractor conveyance, multi-array production logging, and mechanical zonal isolation in an ERD well from a rig up constrained offshore platform using a 90-ft portable mast. It establishes a replicable model for safe, cost-effective rigless diagnostics and zonal remediation in mature offshore assets.

Introduction

Cairn Oil & Gas, Vedanta Ltd. operates the CB/OS-2 block located in the offshore Cambay Basin, off the western coast of India near Gujarat. Within this block lies the Lakshmi Field, discovered in 2000, with commercial production commencing in 2002. Initially developed as a gas-producing asset, the field has since evolved into a major oil and gas producing hub, primarily targeting the hydrocarbon-rich Tarkeshwar sands. The field consists of three structural culminations—LA, LB, and GA—formed during a transgressive depositional phase following the Late Oligocene lowstand unconformity. Two main formations contribute to production: the Babaguru and Tarkeshwar. The Tarkeshwar formation is further subdivided into upper, middle, and lower intervals, with oil currently being produced from the lower Tarkeshwar zones. The reservoirs exhibit high permeability (multi-darcy range), with pressures around 1500 PSIG and temperatures near 85°C. Produced fluids are light and mobile, with viscosities between 0.5-0.6 CP and a solution gasoil ratio of ~500 SCF/STB.

Development of the Lakshmi Field has relied heavily on infill drilling and diverse completion strategies, including various sandface designs. Between 2002 and 2018, numerous directional wells were drilled from the Lakshmi Alpha and Bravo platforms using moderate step-out ratios (0.4-2.0:1) to access targeted formations up to ~1344 meters of total vertical depth. These conventional wells, characterized by low to moderate deviation, were effective in accessing reservoir zones. However, recent development challenges and the need to access untapped reserves farther from surface locations led to a shift in well construction philosophy. The planned well, referred to here as the subject well, represented a step change in Extended Reach Drilling (ERD) complexity, with a 3:1 step-out ratio. This aggressive trajectory introduced significant operational constraints—particularly around tool conveyance, diagnostics, and completion accessibility—that traditional completion and intervention methods were ill-equipped to handle.

Well Trajectory & Completion Overview

The subject well was drilled to a total measured depth (MD) of 3830 meters with a true vertical depth subsea (TVDSS) of 1235 meters. It follows a "Build-Hold" well trajectory, attaining a maximum inclination of 80-84° as shown in Fig 1 and Fig 2, which is sustained over approximately 2,000 meters of measured depth. This trajectory makes it one of the longest reach wells drilled from the platform.

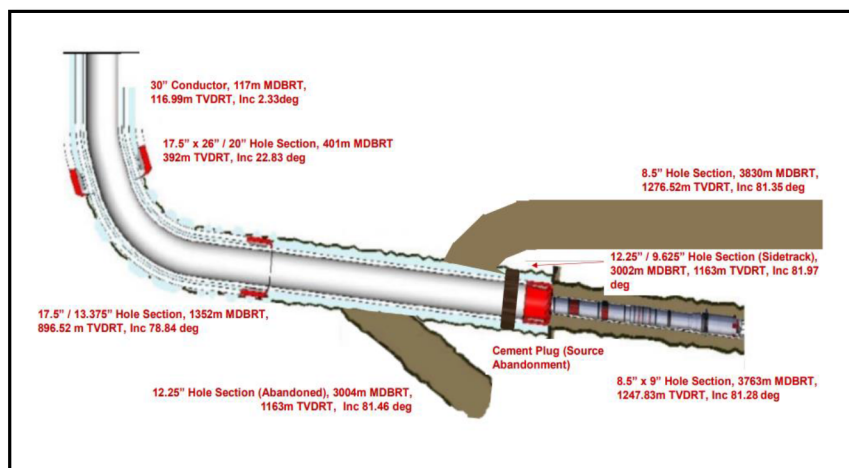


Figure 1—Wellbore Schematic

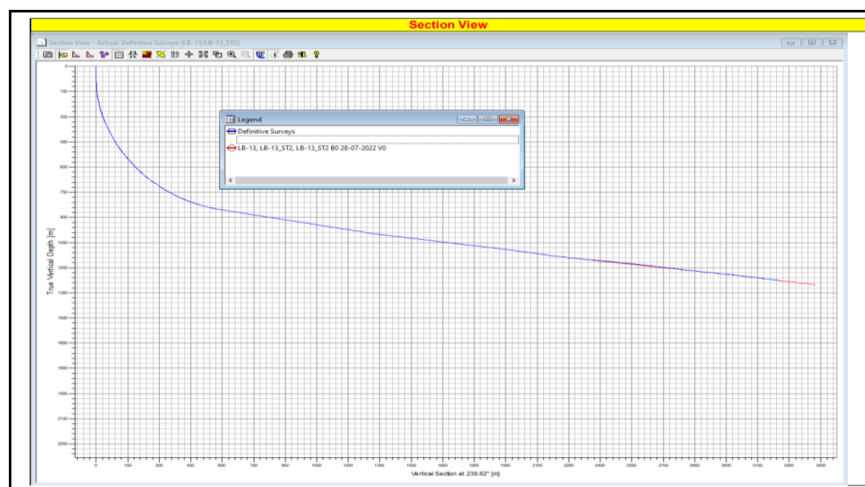


Figure 2—Well Trajectory

The completion design comprises a multi-zone architecture tailored to accommodate both oil and gas production. The lower completion targets the primary oil-bearing and associated gas zones and consists of 5-1/2" tubing deployed with standalone wire-wrapped sand screens (300-micron rating) and nine Sliding Sleeve Doors (SSDs) with a 3.313" internal profile to enable selective stimulation and zonal control. Currently, eight of the nine SSDs are open. It is noteworthy that, due to unforeseen operational contingencies encountered during completion, a coiled tubing (CT) bottomhole assembly fish(SSD manipulation toolstring)was intentionally left in the wellbore between SSD-8 and SSD-9 in the lower completion section. The middle completion targets the T3w shallow gas sand, which is cased and perforated across the 9-5/8" casing and 4.5" tubing. This section is completed using auto gas lift flow valves integrated with sand control screens. It also includes one Gas Lift Mandrel (GLM) with a 1" pocket equipped with a dummy valve and a 3.87" Formation Isolation Valve (FIV). The zone is compartmentalized using two production packers and includes a dedicated Sliding Sleeve Door (SSD) to facilitate zonal isolation and access. The upper completion comprises a 4.5" tubing string incorporating a Sub-Surface Safety Valve (SSSV) with a 3.813" selective profile, five Gas Lift Mandrels (GLMs) with 1.5" pockets equipped with shearable valves, and a circulation Sliding Sleeve Door with a 3.688" profile for operational flexibility.

This complex completion design as shown in Fig 3—spanning multiple reservoir zones with varied flow profiles and intervention constraints—presents considerable challenges for any post-completion well intervention or production optimization activity.

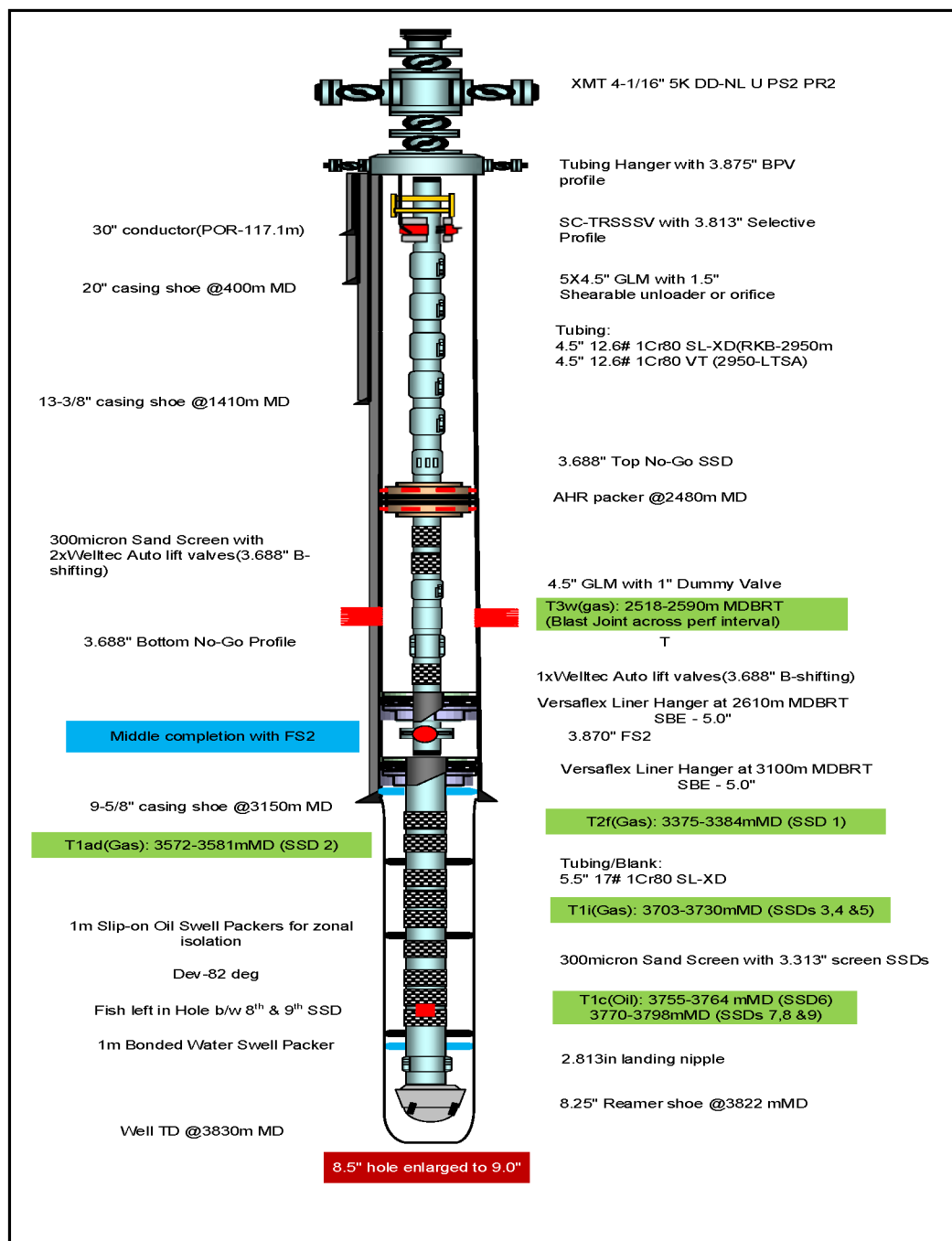


Figure 3—Well Completion Schematic showing all the key jewelry

Well Production Performance & History

The subject well commenced production in September 2022, delivering a strong initial oil rate of approximately 2,100 BOPD with a commingled GOR of ~9,000 SCF/BBL, as illustrated in Figure 4(a). Over the subsequent months, the well exhibited a gradual decline in oil output by ~300 BOPD and a sharper reduction in GOR to approximately 3,200 SCF/BBL. With an assumption that initial gas cap was blown down was completed & hence drop in GOR was expected, a screen backwash operation to address drop in liquid rate was carried out in February 2023, which resulted in a production gain of nearly 500 BOE/day, suggesting successful remediation of formation damage or removal of near-wellbore plugging materials.

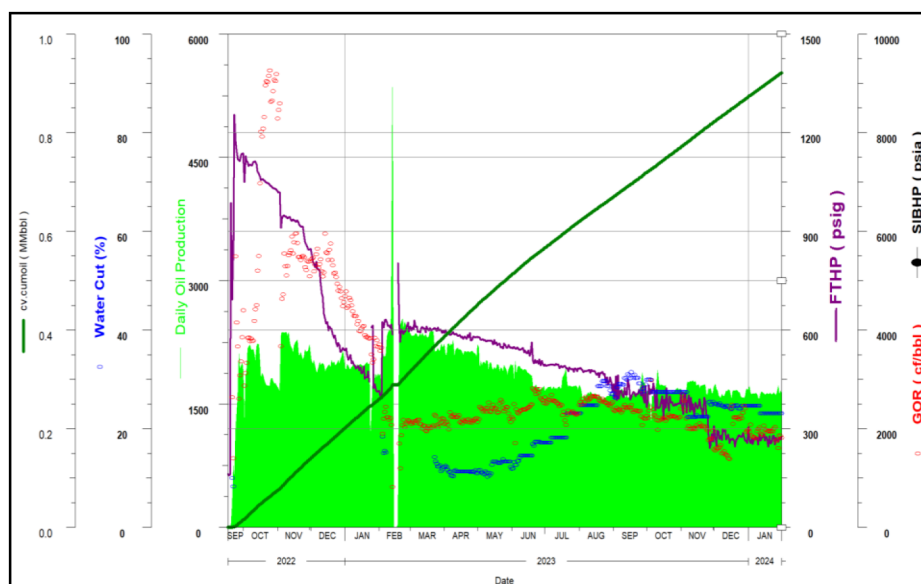


Figure 4a—Production plot from Sept'22 till Jan'24

However, in March 2023, the onset of water production was observed, followed by a steadily increasing water cut over the following months. In an effort to recover declining oil rates, another screen backwash was executed in March 2024, as shown in Figure 4(b). Contrary to expectations, this intervention coincided with a rapid rise in water cut, surpassing 95%, effectively nullifying oil production. Subsequent production optimization attempts—including drawdown enhancement through reduced wellhead pressure and surface gas injection tuning—were undertaken to redistribute inflow and revive oil output. Nevertheless, these measures failed to yield significant improvement. This persistent high water cut suggests the development of a dominant water-producing pathway, either within the openhole horizontal section or due to a water breakthrough from the shallow gas zone into the producing interval.

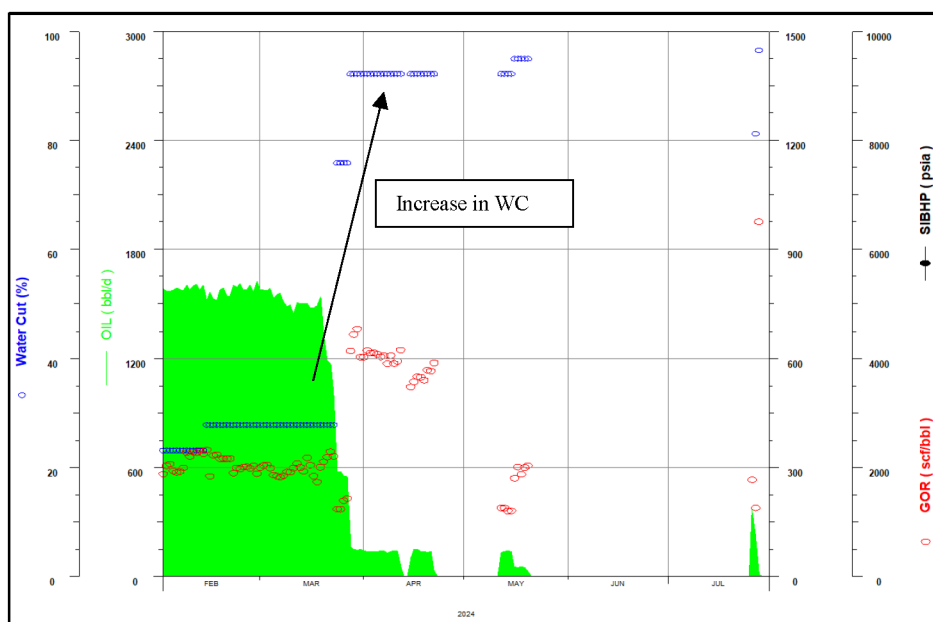


Figure 4b—Production plot showing sudden spike of WC post Mar'24

Intervention Planning, Challenges and Mitigation

The primary objective of this intervention was to diagnose and isolate the dominant water-producing intervals using Eline and tractor- conveyed Production Logging Tool (PLT) combined with selective manipulation of SSDs and AGVs in a highly deviated ERD wellbore. The operation posed several technical challenges due to the complex completion architecture, extended horizontal section, and multiple restriction profiles. structured mitigation plan was developed and implemented to address these constraints, as summarized below.

Challenge 1: Tool Conveyance Across Complex Wellbore Geometry

Conveying the intervention toolstring to the target depth posed a significant challenge due to the well's extended-reach trajectory and multiple ID restrictions. The planned operation required tractoring the toolstring from 950 mMD to 3790 mMD (SSD-8 in the lower completion), a distance of nearly 2900 meters through tortuous wellbore geometry. During the initial original completion movement (OCM), the 3-1/8" OD tractor encountered repeated held ups across the 3.688" ID auto gas lift valves (AGV) and Formation Isolation Valve (FIV). In contrast, summation of criticalities in all the runs & field validation confirmed that toolstring with OD < 2.9" could navigate the AGV & FIV successfully. Moreover, simulation studies revealed that a conventional 2-1/8" tractor would be underpowered for tractoring across the 5-1/2" lower completion, especially considering the extended horizontal section and potential frictional resistance.

Mitigation Strategy. To address these conveyance challenges, the following modifications were implemented:

- A customized 2-1/2" OD slim tractor as shown in [Fig 5\(a\)](#) comprising three modules (OD < 2.9") was deployed, successfully tractoring the toolstring to TD within the subject well.
- A 2-1/8" high-expansion anchor-stroker as shown in [Fig 5\(b\)](#) was employed, capable of anchoring inside 5-1/2" tubing and delivering up to 12,000 lbf of stroking force.
- 2-1/8" high-expansion well keys as shown in [Fig 5\(c\)](#) were integrated to mechanically engage with both the 3.688" AGV and the 3.313" SSD profiles, ensuring proper transmission of stroking forces during manipulation.
- To ensure operational success, an extensive Surface Integration Test (SIT) was conducted at Suvali base using spare Screen SSD and AGV assemblies.



Figure 5a—2-1/2" OD slim tractor wheel section



Figure 5b—2-1/8" Anchor HEX and Stroker Assembly

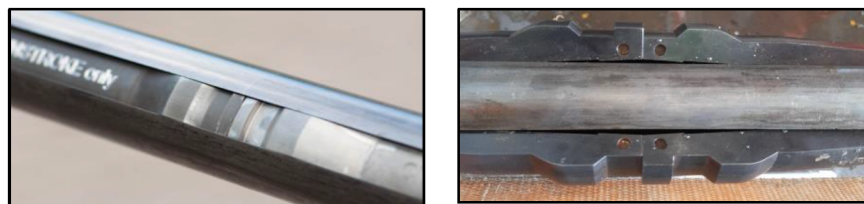


Figure 5(c)—2-1/8" Well Keys HEX - Closed and Expanded

This optimized tractor-stroker-well key configuration in a single BHA proved critical in navigating the multi-restriction well profile and executing the manipulation of sleeves of SSDs effortlessly

Challenge 2: Platform Crane Rig-Up Height Limitation

Tractor-based SSD/AGV manipulation required deploying a tool string approximately 72 feet long, which included three modules of the 2-1/2" tractor along with extended stroker and well key assemblies. Factoring in the E-line pressure control equipment (PCE), the total rig-up height needed was around 90 feet. However, the LB platform crane posed a significant constraint, with a maximum allowable rig-up height of only ~60 feet, rendering conventional deployment infeasible.

Mitigation Strategy.

- To overcome this constraint, it was decided to proceed with a MAST-based deployment to accommodate the required rig-up height without relying on the platform crane. Extensive discussions and evaluations were carried out with multiple vendors capable of providing MAST services, and the vendor with technically closest to our requirement was selected as the service provider for the 90 ft MAST solution.
- A joint site visit was conducted by MAST team and E-line services team to the LB platform. This visit confirmed the physical feasibility of erecting the MAST at the designated corner slot location on the subject well.
- To ensure the structural safety of the platform during operations, a comprehensive structural integrity analysis was performed. Petro6, in coordination with the Department of Ocean Engineering at IIT Madras, evaluated the structural adequacy of the platform under maximum expected load conditions during the MAST deployment.
- Finally, the petroleum engineering and operations teams visited vendors base to physically inspect the MAST assembly and validate its operational readiness for offshore deployment in the constrained platform environment.

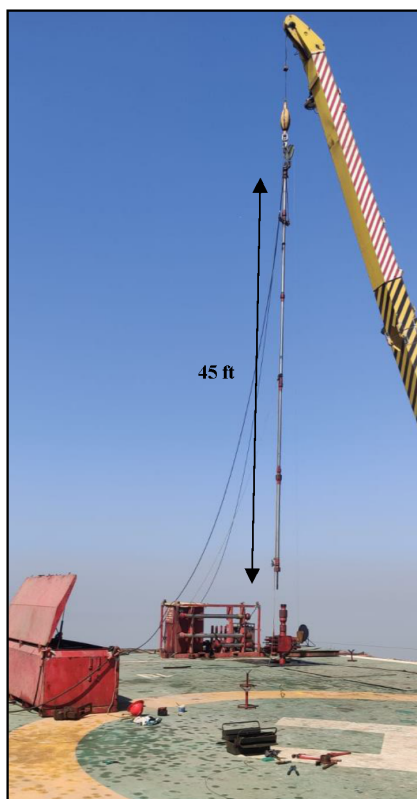


Figure 6(a)—Rig up using platform crane

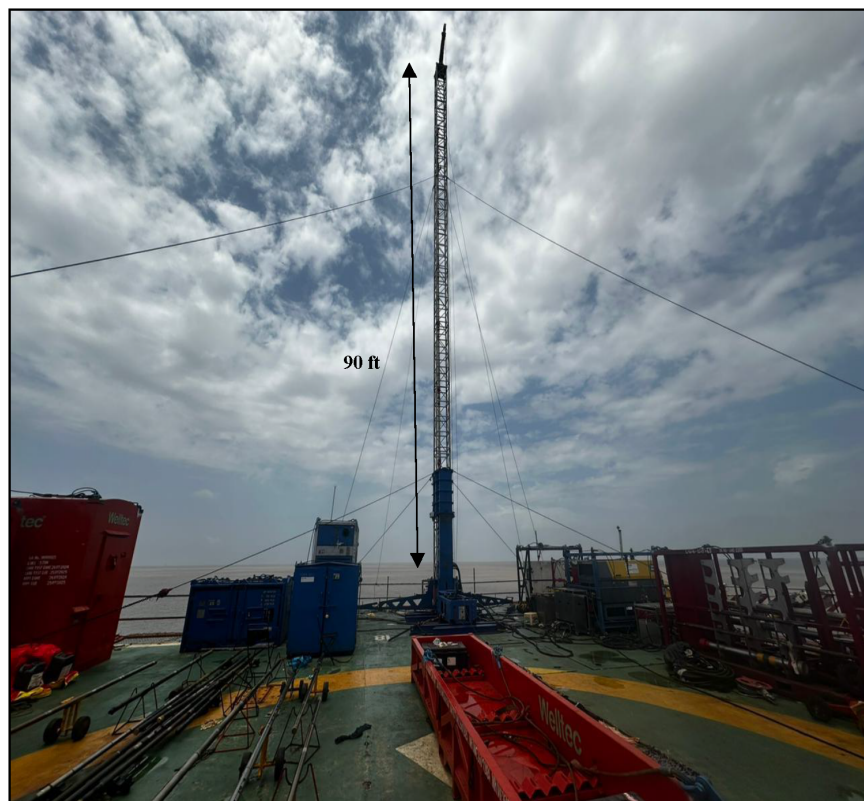


Figure 6(b)—Portable MAST to accommodate 90ft rig-up

Challenge 3: PLT Logging in a Horizontal Wellbore with High Water Cut

Production Logging in horizontal wells poses significant complexity, particularly in high water cut scenarios. One of the primary challenges is phase segregation, where oil and water tend to separate along the horizontal section, making it difficult to pinpoint the exact location of water entry. In the case of subject well, the water cut had exceeded **95%**, further complicating the interpretation of flow profiles and the identification of the active water-producing zones.

Mitigation Strategy.

- To enhance zonal diagnostic resolution in an ERD well, a Multi-Array Production Logging Tool in conjunction with tractor BHA was deployed. This tool integrates capacitance (CAT), resistance (RAT), and spinner (SAT) sensors, allowing for a more comprehensive multiphase flow characterization across the wellbore as shown in [Fig 7](#)
- However, in recognition of the potential limitations of PLT data under such high water cut conditions, a contingency plan was established. In case the logging results proved inconclusive, a trial-and-error based selective sleeve closure strategy was pre-planned, wherein SSDs and AGV in suspected zones would be sequentially manipulated to observe changes in water cut and production response—thereby narrowing down the water entry interval through elimination.



Figure 7(a)—Spinner Array Tool (SAT sensors)



Figure 7(b)—Capacitance Array Tool (CAT sensors)

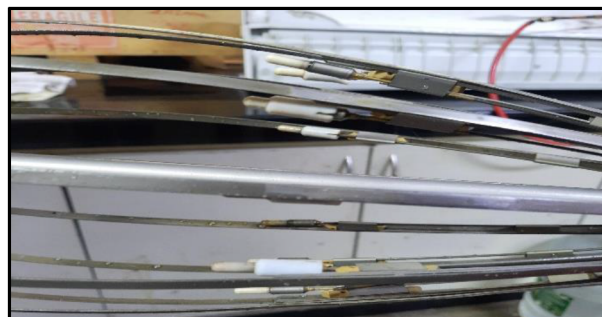


Figure 7(c)—Resistivity Array Tool (RAT sensors)

Challenge 4: Offshore Logistical and Weather-Related Constraints

The intervention was executed under critical offshore conditions, which significantly impacted routine logistics such as marine transfers for personnel and daily utilities. Additionally, the helideck remained non-operational for the duration of the campaign, further complicating emergency preparedness and personnel movement. The platform was temporarily manned with approximately 35 personnel across multiple service partners, necessitating extensive planning for resource management and contingency support.

Mitigation and Risk Management Strategy.

- A comprehensive risk assessment and contingency plan were formulated post cumulative discussions with all the stakeholders in advance to ensure operational continuity and safety.
- Critical arrangements were made for offshore accommodation, food, potable water, power, lighting, and medical supplies to support extended manned operations.
- A helideck spotting protocol was established for emergency scenarios. Marine support infrastructure, including standby tugboats and vessels, was deployed throughout the campaign. Furthermore, approvals were obtained from near by Port authorities to facilitate emergency evacuation, if necessary.

These pre-emptive measures ensured uninterrupted execution of the complex intervention despite logistical and weather-related challenges.

Execution and Results

A successful dummy PLT run confirmed tool conveyance across the auto gas lift valve (AGV) located in the middle completion, with a clear tag obtained at 3724 mMD—positioned between SSD-4 and SSD-5. Subsequently, production logging passes were conducted across multiple reservoir intervals: T3w gas-bearing zone in the middle completion, T2f and T1ad gas zones in the lower completion, and the T1i oil-bearing zone up to SSD-4. These passes were carried out at varying tool speeds of 10 m/min, 20 m/min, and 6.5 m/min to ensure data integrity and dynamic response assessment.

The PLT run involved approximately 6.8 km of tractoring, successfully executed in a single operation, enabling comprehensive coverage of the targeted intervals. Below SSD-4, the predominant fluid production was water with only traces of hydrocarbons. Based on Resistivity Array Tool (RAT) measurements, SSD-3 was identified as the primary contributing interval, with SSD-4 showing minor contribution. In the upper sections, RAT responses indicated minor hydrocarbon presence above SSD-2. Simultaneously, Spinner Array Tool (SAT) data suggested fluid intake through SSD-2, evidenced by a noticeable reduction in spinner RPM above this interval — implying backflow or crossflow from the wellbore to SSD-2. Furthermore, data from AGV-2 across T3w zone showed a marked reduction in temperature response and an increase in spinner velocities, which collectively pointed to gas influx originating from this zone.

Following the PLT operation, a stoker-based shifting intervention was successfully executed to close SSD-1 and SSD-2, which were identified as potential thief zones and were responsible for crossflow of hydrocarbon from bottom zones to shallower zones across SSD1 & SSD2. The stoker applied a controlled downward force in the range of 4000 to 6000 lbf as depicted in Fig 9(a) & Fig 9(b), achieving positive shifting and closure of both sleeves. The operation confirmed mechanical responsiveness of the sleeves and validated the capability of the tractor-conveyed stoker system to perform zonal isolation even in a high-deviation ERD well environment.

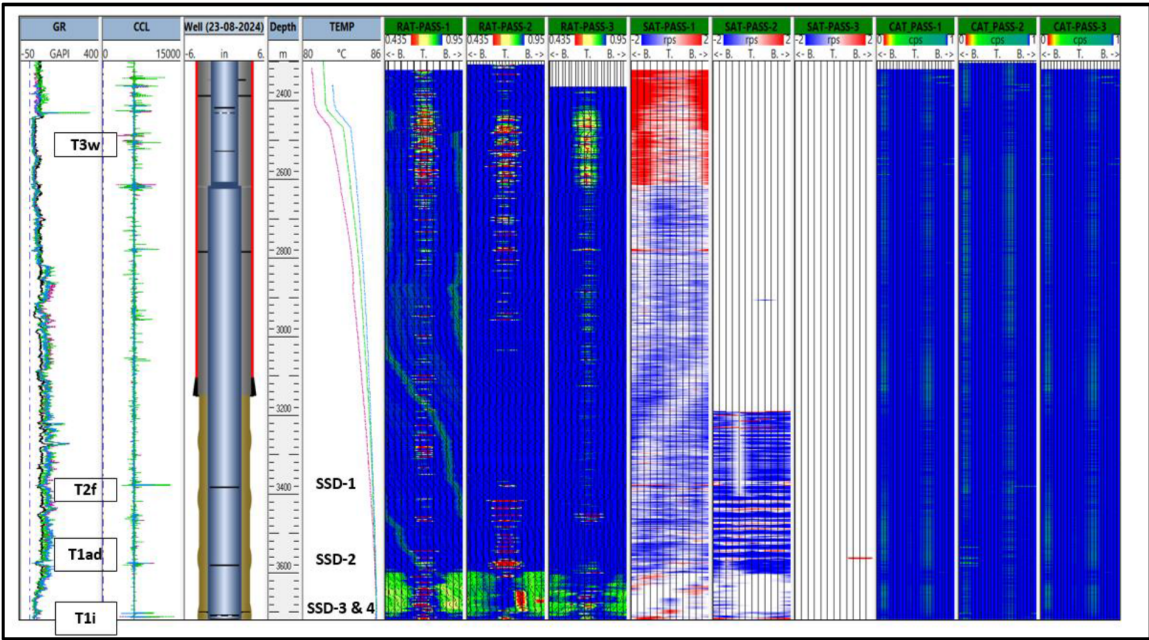


Figure 8—Production Logging Interpretation Plot

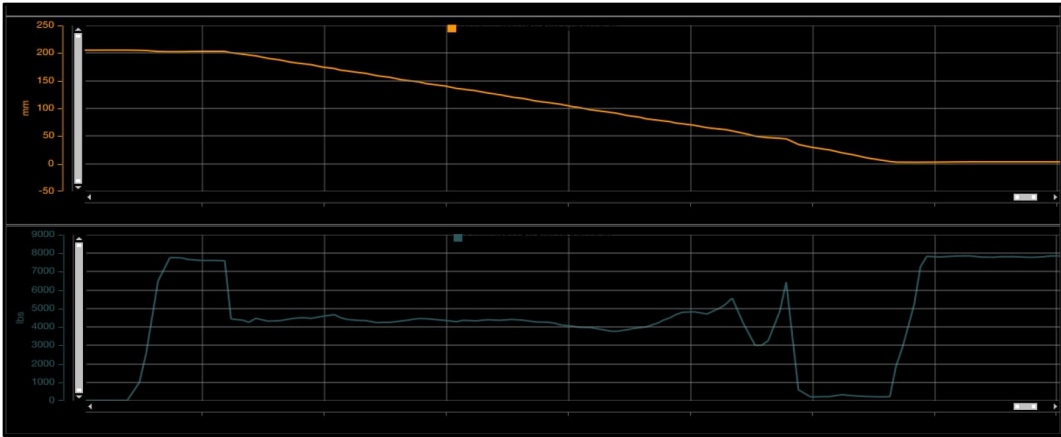


Figure 9(a)—Stroking Force and Stroke Length - SSD 1 Closing

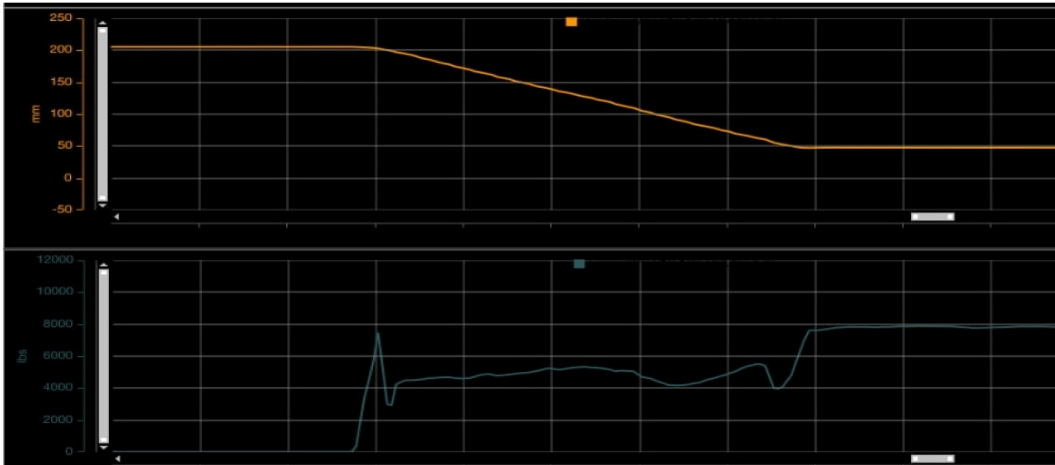


Figure 9(b)—Stroking Force and Stroke Length - SSD 2 Closing

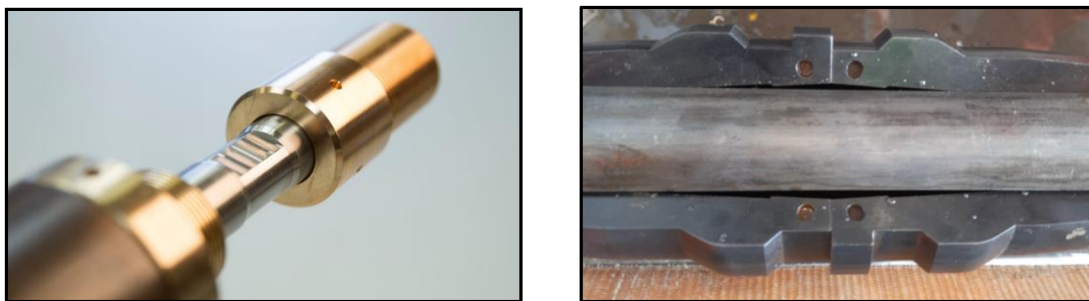


Figure 9(c)—2-1/8" HEX Stroker and HEX keys

Well Activation post intervention

This section presents a comprehensive account of the well activation strategy implemented for the subject well, with specific emphasis on surface production response to choke manipulations and artificial lift support. The evaluation is based on trends observed in surface parameters, as depicted in Fig 10 (Activation Plot). The primary objective was to achieve stabilized production with reduced water cut through controlled bean-up and gas lift optimization during the post-intervention flowback phase.

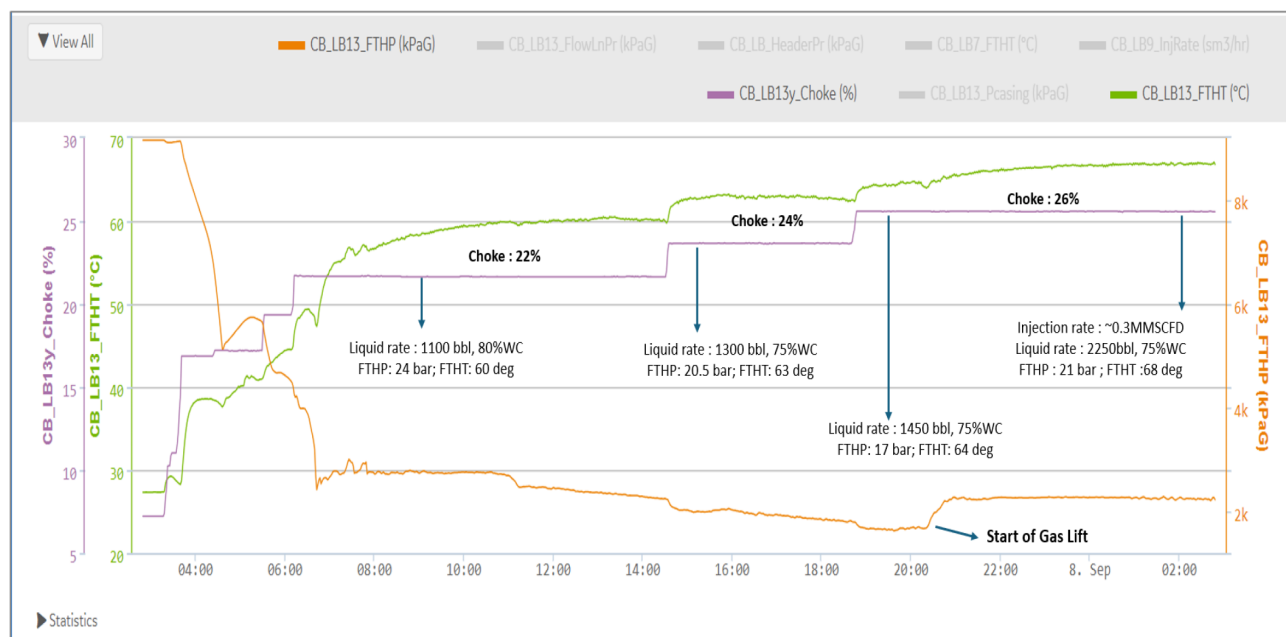


Figure 10—Wellhead parameters during activation phase showing choke manipulations and stabilized flow rates.

Activation Sequence and Observations.

- The well was systematically beaned up in incremental steps, beginning from a fully shut-in condition. The choke was initially opened up to 22%, which corresponds to the last known stable operating condition prior to the well shut-in. At this setting, the well began to flow naturally, stabilizing at a Flowing Tubing Head Pressure (FTHP) of approximately 24 bar, delivering a net liquid rate of ~1100 BLPD with an associated water cut of ~80%.
- As production continued, a gradual decline in FTHP was observed, indicating possible wellbore loading or reservoir drawdown. To sustain flow and enhance drawdown efficiency, the choke was further opened to 24%, resulting in an increase in liquid rate to ~1300 BLPD. Notably, this step also yielded a reduction in water cut to ~75%, suggesting effective clean-up of the near-wellbore region and improved hydrocarbon contribution.

- In the final stage of the activation, the choke was increased to 26%, and gas lift was initiated at ~0.3 MMSCFD. The gas injection facilitated a reduction in hydrostatic head and improved phase segregation within the wellbore, effectively supporting the lifting of reservoir fluids. Post gas lift initiation, the well stabilized at FTHP of ~21 bar, with a significant ramp-up in liquid production to ~2250 BLPD, while maintaining a water cut of ~75%.

Conclusion, Results, and Key Learnings

The successful execution of a rigless well intervention in the extended-reach horizontal well demonstrated the technical feasibility and operational excellence required to identify and isolate high water-cut producing zones through tractor-conveyed PLT logging and SSD/AGV manipulation.

Despite a constrained offshore platform environment, critical weather conditions, and complex well geometry, the intervention was delivered safely and effectively. A robust combination of fit-for-purpose toolstring design, detailed pre-job planning including SITs and simulations, and proactive contingency management enabled flawless execution.

Key Results

- The well was restored to a stable production of ~590 BOPD, significantly improving from >95% water cut scenario.
- Downhole Diagnostics: Multi-array PLT conveyed via tractor provided crucial zonal conformance data. SSD-2 was identified as the primary thief zone of hydrocarbon from bottom zones, supported by capacitance/resistance data and flow responses.
- Zonal Isolation: SSD-1 and SSD-2 were successfully closed using the stoker tool at ~4000-6000 lbf, leading to shutoff of thief zones & substantial reduction in water contribution from lower completion zones.
- Enhanced Production Performance: Post-intervention activation resulted in ~2250 BLPD with a reduced water cut of ~75% and stabilized flowing tubing head pressure of 21 bar, validating the effectiveness of zonal isolation.
- Record-Length Tractoring: A single-run tractor operation of ~6.8 km in an ERD well set a new benchmark in the field for extended horizontal conveyance.
- Safe and Timely Delivery: The job was executed safely within a tight operational window during critical offshore conditions, facilitated by a 90-ft MAST rig-up, platform structural evaluations, and field-readiness preparations.

Key Learnings

- Multi-Sensor Logging is Crucial in High WC Environments: Multi Array PLT toolstring with spinner, capacitance, and resistance sensors provide reliable data in horizontal wells with 90%+ water cut to discern zonal contribution effectively.
- Customized Toolstring Design Enhances Efficiency: Use of 2-1/2" modular tractors, compact high-expansion stoker, and purpose-built well key configurations enabled seamless conveyance and actuation across multiple SSDs and AGV.
- Pre-Job Simulation and SITs are Non-Negotiable: Conducting detailed tractoring simulations and physical SITs with spare SSD/AGV inventory proved indispensable in reducing uncertainties, estimating exact BHA length and ensuring job execution smoothly.
- MAST Deployed as a Platform Enabler: Structural rig-up limitations were overcome through the integration of a 90-ft self-jacking MAST, expanding the intervention envelope for high-deviation ERD wells from constrained unmanned platforms.

- Offshore Readiness during critical weather conditions is achievable: With comprehensive risk assessments, evacuation strategies, and logistic preparedness, critical operations can be delivered safely even under weather and access constraints.

Acknowledgement

The authors express their sincere gratitude to all team members who contributed to the successful planning, execution, and documentation of this complex intervention. Special thanks are extended to our esteemed service partners for their technical expertise and dedicated support throughout the project. The unwavering guidance and encouragement from the leadership team were instrumental in navigating challenges and achieving a safe and effective outcome.

Nomenclature

BOPD	-barrels of oil per day
MMSCFD	-Million standard cubic feet per day
BLPD	-barrels liquid per day
PSIG	-pounds per square inch gauge
CP	-centipoise
”	-inches
SCF/BBL	-standard cubic feet/ barrel
%	-percentage
ERD	-extended reach drilling
OHSAS	-open hole standalone screens
SSD	-sliding sleeve door
AGV	-auto gas lift valve
PLT	-production logging tool
MAPLT	-multi array production tool
NPT	-non productive time
HEX	-high expansion
SIT	-surface integrity test
FHTP	-flowing tubing head pressure

References

1. Abdalla, A.A., Jones, M.R., and Grayson, T.D. 2018. Comprehensive Review of Well Tractor Technology in Highly Extended-Reach Wells. Presented at the SPE/ICoTA Well Intervention Conference and Exhibition, The Woodlands, Texas, 27-28 March. SPE-189590-MS.
2. Al-Yami, A.S., Aziz, K., and Economides, M.J. 2007. Production Logging in Difficult Well Configurations. Presented at the International Petroleum Technology Conference, Dubai, UAE, 4-6 December. IPTC-11265-MS. <https://doi.org/10.2523/IPTC-11265-MS>
3. Ali, S.A., Marhoon, A., and Ogunsanya, A. 2008. A Novel Approach to Production Logging in Multiphase Horizontal Wells. *JPT- Journal of Petroleum Technology* **60** (11): 86–89. SPE-114995-PA.
4. Burmistrova, A., Mavko, B., and Abakarov, A. 2023. Extended-Reach Drilling (ERD)—The Main Problems and Current Achievements. *Applied Sciences* **13** (7): 4112.
5. Bhosale, S., Misra, A., Ranjan, A. 2024. Navigating Complexity: Planning and Execution of the Longest ERD Well with Open Hole Completion Design in India's West Coast Cambay Offshore

Field. Paper presented at the Offshore Technology Conference Asia, Kuala Lumpur, Malaysia, 27-29 February. OTC-34677-MS.

6. Matharu, J., Squires, S., and Watson, D. 2020. Innovative Wireline Mast Design Improves Safety and Performance for Standalone Operations. Paper presented at the SPE/ICoTA Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 24-25 March. SPE-199831-MS.